

Possession Arrangements

Revision Guide

Module 4 | GERT8000-T3 (Iss 13, Dec 2025) | EMR PPT 30-03

1. What is a Possession?

A possession is taken when engineering work requires the complete stoppage of all normal train movements on a running line, and engineering train movements are needed in connection with that work.

- Normal signalling is suspended on a line under possession
- The line is protected by signals held at danger, detonators and Possession Limit Boards (PLBs)
- Except in urgent situations, all possession details must be published in the Weekly Operating Notice (WON) or Engineering Notice (EN)
- Limits of a possession can be shortened or lengthened if published in the WON/EN, or agreed by Operations Control in exceptional circumstances

2. Key Personnel

Role	Responsibility
PICOP (Person in Charge of Possession)	Overall authority over the entire possession. Authorises all movements between detonator protection and work sites, between work sites, and leaving work sites. Wears yellow armlet/badge: PERSON I.C. POSSESSION in red letters.
ES / SWL (Engineering Supervisor / Safe Work Leader)	In charge of each individual work site. Authorises all movements entering a work site and within the work site. Wears yellow armlet/badge: ENGINEERING SUPERVISOR in blue letters.
Competent Person (CP)	Can pass on instructions from the PICOP or ES/SWL to the driver. Cannot authorise beyond the limits given by the PICOP or ES.
Signaller	Authorises movements towards and away from the possession. Grants the possession to the PICOP. Keeps protecting signals at danger throughout.

3. The T3 Possession Pack

Before the possession, you should receive a copy of the Possession Pack (also called the T3 Pack or Ballast Circular). It includes the following details relevant to a driver:

- The nature and location of the work
- The access point and route to site (for those walking)
- The location of the work site marker boards (WSMBs)
- Communication details
- Any known hazards (e.g. limited clearance)

4. Possession Protection

Protecting Signals

- All controlled signals within the possession and all signals leading to the possession are placed and maintained at danger
- Points are set to protect the possession

Detonator Protection

Detonator protection consists of:

**3 detonators placed on the SAME rail, 20 metres apart
+ a Possession Limit Board (PLB) placed in the "four-foot" adjacent to the MIDDLE detonator**
**Standard distance: normally placed 400 metres beyond the protecting signal
or 400 metres beyond any points — whichever is further**
Placed at EACH END of the possession and on any line joining at a junction
NOT placed where there are loops or crossovers between running lines

Single Line — Token Alternative

- On a single line, the PICOP may take the token as protection instead of detonator protection
- In this case, no detonators are placed for the possession
- The signaller and PICOP agree the exact location the driver should proceed to

5. Work Sites and Work-Site Marker Boards (WSMBs)

- Each work site is protected by WSMBs placed in the "four-foot" at each end
- One work site's WSMB must be no closer than 100 metres from the WSMB of the next work site
- WSMBs are NOT needed if there will be no engineering trains or OTM movements in the possession

WSMB showing TWO RED FLASHING LIGHTS = Entrance to work site

→ Only the ES / SWL can authorise you to pass this and enter the work site

WSMB showing TWO YELLOW FLASHING LIGHTS = Exit from work site

→ Only the PICOP can authorise you to pass this and leave the work site

6. Authority for Movements — Who Authorises What?

You must only make a movement if you have the authority of the correct person. Use the table below as a guide.

Movement	Who Authorises	Notes
Proceed from either end towards the detonator protection	SIGNALLER (personally)	Signaller must get PICOP permission first
Proceed towards possession meeting point (token line — entering)	SIGNALLER (personally)	Agreed location with PICOP
Enter possession at an intermediate point (be met there)	SIGNALLER (personally)	You will be met and given instructions at the intermediate point
Past the detonator protection INTO the possession	PICOP (or CP on their behalf)	
Through protecting points at intermediate entry point	PICOP (or CP on their behalf)	
Between detonator protection and nearest work site	PICOP (or CP on their behalf)	
Enter/leave from adjacent siding under possession	PICOP (or CP on their behalf)	
Pass WSMB (two yellow lights) — leaving work site	PICOP (or CP on their behalf)	
Between work sites	PICOP (or CP on their behalf)	
Pass WSMB (two red lights) — entering work site	ES / SWL (or CP on their behalf)	
All movements within a work site	ES / SWL (or CP on their behalf)	
Pass through protecting points when LEAVING at intermediate point	SIGNALLER (personally)	
Proceed past detonators when LEAVING the possession	SIGNALLER (personally)	Detonator protection removed first

7. Driver Duties — Key Rules

Before Entering

Temporarily isolate TPWS before entering the possession.

Speed

Location	Maximum Speed
Anywhere within the possession (entering, moving, leaving)	25 mph
Within a work site	5 mph (unless ES/SWL gives specific instructions)
At all times	Proceed at caution — be prepared to stop before any handsignal

During the Movement

- **Reach a clear understanding with the authorising person — what to do and how far to go**
- Sound the horn when starting a movement within a possession (TW1 rule)
- Use GSM-R radio at any time to discuss details of the movement
- If detained outside a work site: red light at BOTH ends, headlights OFF

Signals Within the Possession

Signal Situation	What to Do
Signal at danger — OUTSIDE a work site	Must NOT pass without PICOP authority (or CP on PICOP's behalf)
Signal at danger — INSIDE a work site	Must NOT pass without ES / SWL authority (or CP on their behalf)
Signal showing PROCEED aspect	May pass without authority — but DISREGARD the aspect; proceed at caution

8. Level Crossings Within the Possession

You must NOT pass over any level crossing unless you have been given instructions to do so.

Authority comes from whoever controls the area you are in (PICOP, ES, or signaller).

Crossing Type	What is Required
AHBC (locally controlled)	Green handsignal from crossing attendant
CCTV / OD / RC (locally controlled)	Green handsignal from crossing attendant
AHBC (NOT locally controlled)	PICOP/ES gets signaller's permission; proceed on instructions given
All other crossings	Follow instructions from person in control of your area; standard crossing rules in S5/TW7 still apply

9. Entering the Possession

Standard Entry — via Detonator Protection

1	Signaller authorises you to pass the protecting signal at danger and proceed to the detonator protection (max 25mph, at caution)
2	You arrive at the detonator protection — this is the PICOP's authority limit boundary
3	PICOP (or CP) authorises you to pass through the detonators into the possession
4	Continue at caution (max 25mph) toward the work site
5	PICOP (or CP) authorises onward movement to the WSMB
6	ES / SWL authorises you to pass the WSMB (two red lights) and enter the work site (max 5mph)

Intermediate Point Entry

- Signaller personally authorises you to proceed to the intermediate point and tells you to stop and get instructions there
- You will be met at the intermediate point by the PICOP or a CP
- Do not pass beyond the intermediate point without being met
- Check any points are correctly set once given permission to proceed after being met

10. Leaving the Possession

1	PICOP authorises you to proceed from the work site toward the detonator protection (max 25mph)
2	You arrive at the Possession Limit Boards — this marks the limit of the PICOP's authority
3	Contact the signaller and wait
4	Only the SIGNALLER can authorise you to leave the possession — past the

	detonator protection
5	PICOP or CP removes the detonator protection to allow the movement
6	Signaller gives clear instructions on where the movement may proceed to
7	If the next signal shows a proceed aspect, signaller tells you whether to obey it

Risk Check

- Check the WON to confirm the published end-of-possession location matches the Possession Limit Board you find on the ground
- If the PICOP has the token (single line), the PICOP tells you to get permission from the signaller to leave

11. Possession Taken or Given Up Around Your Train

Possession Taken Around Your Train

- Must be published in the WON or Engineering Notices
- You are signalled normally to the specified signal, block marker or FTAP
- Signaller tells you: make no further movement until the PICOP or ES authorises you to move

Possession Given Up Around Your Train

- Does NOT need to be published
- Movements after the possession is given up MUST be in the signalled direction and driven from the front end (leading cab)

Two scenarios apply:

- If your train is within a work site: the ES/SWL tells you the work site is being given up — do not move your train
- If your train is NOT within a work site: the PICOP tells you the location and identity of the signal or block marker you must stop at

At the specified signal or block marker:

- Immediately contact the signaller
- Make no further movement until the signaller tells you to proceed
- PICOP will tell you the line is no longer to be considered under possession
- Signaller will tell you the conditions under which you may proceed

12. Protection Zones (PZ)

A Protection Zone (PZ) is an alternative to a full possession. It can only be set up when details are published in the WON or EN.

When a PZ Can Be Used

- Only one engineering train AND one OTM required
- Details must be published in the WON or Engineering Notice
- Set up between the ES/SWL and the signaller
- A protecting signal is kept at danger; exit signal beyond the PZ is also kept at danger
- Additional protection: T-COD (track circuit operating device) or disconnection of signalling equipment

PZ Distances

Boundary	Distance
Protecting signal to start of PZ	At least 400 metres beyond protecting signal
End of PZ to exit signal	At least 200 metres before exit signal
Wrong-direction movements	Must NOT bring the train within 400 metres of the protecting signal

Driver Duties in a PZ

Stage	What happens
Entering the PZ	Signaller authorises past protecting signal to a specified signal/FTAP. ES/SWL then takes over authority.
Within the PZ	Only ES/SWL authorises movements. Make all movements at caution. Follow ES/SWL instructions for signals at danger and any wrong-direction movements.
Leaving the PZ	ES/SWL tells you to move to end of PZ (min 200m before exit signal). Stop and contact the signaller. Signaller authorises you to proceed normally, obeying all signals.

13. Key Numbers at a Glance

Rule	Value
Maximum speed throughout a possession	25 mph
Maximum speed within a work site	5 mph (unless ES/SWL specifies otherwise)
Standard detonator protection distance	400 m beyond protecting signal or points
Number of detonators in detonator protection	3

Spacing between detonators	20 metres apart, same rail
PLB position	Adjacent to the MIDDLE detonator, in the "four-foot"
Minimum gap between WSMBs of adjacent work sites	100 metres
PZ start — minimum distance beyond protecting signal	400 metres
PZ end — minimum distance before exit signal	200 metres
Wrong-direction movements in PZ — min distance from protecting signal	400 metres
Headlights when detained outside work site	OFF — red light at both ends instead

Q & A — Possession Arrangements

All questions sourced from GERT8000-T3 Issue 13 (December 2025) and EMR PPT 30-03.

Q & A
Q: What is a possession and why is one taken? A: A possession is taken when engineering work requires the complete stoppage of all normal train movements on a running line and engineering train movements are needed. Normal signalling is suspended within the possession.
Q: Who is the PICOP and what are their main responsibilities? A: The PICOP (Person in Charge of Possession) has overall authority over the possession. They authorise all movements between the detonator protection and work sites, between work sites, and leaving work sites. They agree the possession details with the signaller.
Q: What does WSMB stand for and what do the two different light displays mean for a driver? A: Work-Site Marker Board. Two RED flashing lights = entrance to work site (only ES/SWL can authorise entry). Two YELLOW flashing lights = exit from work site (only PICOP can authorise departure).
Q: What is detonator protection and how is it set out? A: Three detonators placed on the same rail, 20 metres apart, with a Possession Limit Board (PLB) placed in the "four-foot" adjacent to the middle detonator. Normally placed 400 metres beyond the protecting signal (or beyond any points).
Q: What is the maximum speed anywhere in a possession, and what is the limit inside a work site? A: Maximum 25 mph throughout the possession. Maximum 5 mph inside a work site (unless the ES/SWL gives specific instructions for a different speed).
Q: What must you do with TPWS before entering the possession? A: You must temporarily isolate TPWS before entering the possession.
Q: A signal within the possession is showing a proceed aspect. Can you pass it and what must you do? A: Yes, you can pass a proceed-aspect signal without authority, but you must disregard the normal meaning of that signal and continue to proceed at caution.
Q: Who is the ONLY person who can authorise you to leave the possession past the detonator protection? A: The signaller — and only the signaller — can authorise you to proceed beyond the detonator protection when leaving the possession.
Q: You are detained outside a work site. What must you do with the headlights and tail lamps? A: Switch the headlights OFF and make sure a red light is showing at both ends of the train.
Q: What rule applies when starting a movement within a possession?

A: You must sound the horn when starting your train when making a movement within a possession (TW1).

Q: There is an AHBC level crossing within the possession that is being locally controlled. What must you see before passing over it?

A: A green handsignal from the crossing attendant.

Q: A possession is to be given up around your train while you are stopped within a work site. Who tells you and what must you do?

A: The ES/SWL tells you that the work site is being given up. You must not move your train. You then wait to be told the possession has been given up by the PICOP, and the signaller will confirm and give conditions for proceeding.

Q: When a possession is given up around an engineering train, what two conditions must apply to the subsequent movement?

A: 1. The movement must be in the normal signalled direction. 2. The movement must be driven from the leading cab (front end).

Q: What is a Protection Zone (PZ) and when can it be used instead of a full possession?

A: A PZ is a smaller, simpler alternative to a possession for engineering work requiring only one engineering train and one OTM. It must be published in the WON or Engineering Notice and is set up between the ES/SWL and the signaller.

Q: When making a wrong-direction movement within a PZ, what distance must you maintain from the protecting signal?

A: The movement must not bring the train within 400 metres of the protecting signal.

Q: How do you leave a Protection Zone?

A: The ES/SWL instructs you to move to the end of the PZ (at least 200 metres before the exit signal), stop, and contact the signaller. The signaller then authorises you to leave the PZ, proceed normally and obey all signals.